

Ime in priimek:

REŠITVE

Vpisna ali ID številka:

Smer študija: IPT UN

1. Reši podano rekurzijo.

$$a_{n+3} - 7a_{n+2} + 24a_{n+1} - 18a_n = 13$$

$$\textcircled{H} \quad x^3 - 7x^2 + 24x - 18 = 0$$

$$x_1 = 1$$

1	-7	24	-18	
1		1	-6	18
1	-6	18	0	

$$x^2 - 6x + 18 = 0$$

$$D = 36 - 4 \cdot 18 = -36$$

$$\sqrt{D} = 6i$$

$$x_{2,3} = \frac{6 \pm 6i}{2} \rightarrow x_2 = 3 + 3i \quad \begin{matrix} a=3 \\ b=3 \end{matrix}$$

$$\rightarrow x_3 = 3 - 3i$$

$$r = \sqrt{3^2 + 3^2} = \sqrt{18} = 3\sqrt{2}$$

$$\varphi = \arctan 1 = \frac{\pi}{4}$$

$$a_n^{(h)} = A \cdot 1^n + (3\sqrt{2})^n \cdot (B \cdot \sin \frac{\pi n}{4} + C \cdot \cos \frac{\pi n}{4})$$

$$a_n = A + (3\sqrt{2})^n \cdot (B \cdot \sin \frac{\pi n}{4} + C \cdot \cos \frac{\pi n}{4}) + n$$

$$\textcircled{P} \quad f(n) = 13 \cdot 1^n \quad \begin{matrix} a=1 \\ l=0 \end{matrix}$$

$$a_n^{(p)} = n^1 \cdot D \cdot 1^n = Dn = 1 \cdot n = n$$

$$D(n+3) - 7D(n+2) + 24D(n+1) - 18Dn = 13$$

$$Dn + 3D - 7Dn - 14D + 24Dn + 24D - 18Dn = 13$$

$$13D = 13$$

$$\underline{\underline{D=1}}$$

2. Z uporabo kitajskega izreka o ostankih reši podani sistem linearnih kongruenčnih enačb.

$$\textcircled{1} \quad 3x \equiv 4 \pmod{5}$$

$$\textcircled{2} \quad x \equiv 8^{456} \pmod{3}$$

$$\textcircled{3} \quad -3x \equiv 1 \pmod{35}$$

$$\begin{aligned} \textcircled{1} \quad 3x &\equiv 4 \pmod{5} \\ 3x &\equiv 9 \pmod{5} \quad |:3 \\ x &\equiv 3 \pmod{5} \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad 8^{456} &\equiv (-1)^{456} = 1 \\ x &\equiv 1 \pmod{3} \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad -3x &\equiv 1 \pmod{7} \quad \text{in} \quad -3x \equiv 1 \pmod{5} \\ 4x &\equiv 8 \pmod{7} \quad |:4 & -3x &\equiv 6 \pmod{5} \quad |:(-3) \\ x &\equiv 2 \pmod{7} & x &\equiv -2 \pmod{5} \\ & & x &\equiv 3 \pmod{5} \end{aligned}$$

$$\begin{array}{llll} c_1 = 3 & m_1 = 5 & N_1 = 21 & x_1 = 1 \\ c_2 = 1 & m_2 = 3 & N_2 = 35 & x_2 = -1 \\ c_3 = 2 & m_3 = 7 & N_3 = 15 & x_3 = 1 \end{array}$$

$$\begin{array}{lll} 21x_1 \equiv 1 \pmod{5} & 35x_2 \equiv 1 \pmod{3} & 15x_3 \equiv 1 \pmod{7} \\ x_1 \equiv 1 \pmod{5} & -x_2 \equiv 1 \pmod{3} \quad |:(-1) & x_3 \equiv 1 \pmod{7} \\ & x_2 \equiv -1 \pmod{3} & \end{array}$$

$$\begin{aligned} x &\equiv (3 \cdot 21 \cdot 1 + 1 \cdot 35 \cdot (-1) + 2 \cdot 15 \cdot 1) \pmod{5 \cdot 3 \cdot 7} \\ x &\equiv 58 \pmod{105} \end{aligned}$$

3. Na množici $\mathbb{N}_0 \times \mathbb{N}_0$ je podana relacija R kot:

$$(x, y) R (a, b) \iff x \cdot y \geq a \cdot b - |a|$$

- (a) Ali je relacija R refleksivna?
- (b) Ali je relacija R simetrična?
- (c) Ali je relacija R antisimetrična?
- (d) Ali je relacija R tranzitivna?
- (e) Ali je relacija R intranzitivna?

Vse odgovore utemelji!

(a) DA ; $\forall (x, y) \in \mathbb{N}_0 \times \mathbb{N}_0 : (x, y) R (x, y)$

$$\begin{aligned} x \cdot y &\geq x \cdot y - |x| \\ |x| &\geq 0 \quad \checkmark \end{aligned}$$

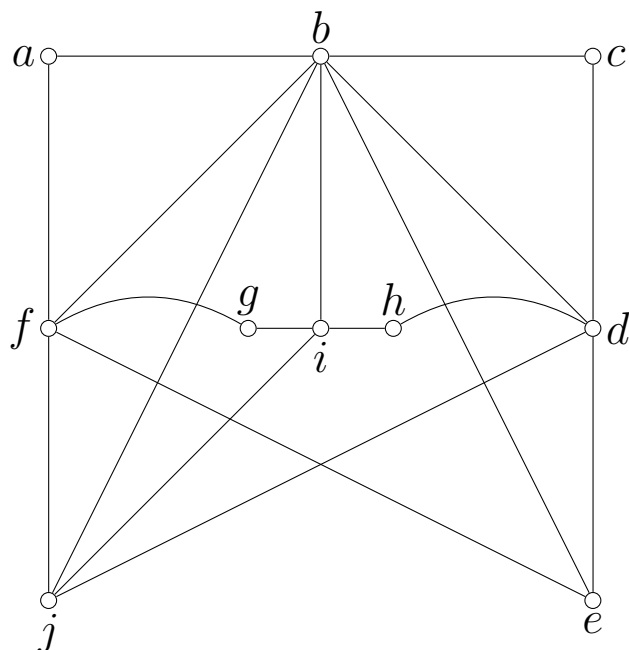
(b) NE ; $(1, 2) R (0, 1)$
 $(0, 1) \not R (1, 2)$

(c) NE ; $(2, 3) R (2, 4)$
 $(2, 4) R (2, 3)$

(d) NE ; $(4, 1) R (2, 3)$
 $(2, 3) R (2, 4)$
 $(4, 1) \not R (2, 4)$

(e) NE ; $(2, 3) R (2, 4)$
 $(2, 4) R (2, 3)$
 $(2, 3) R (2, 3)$

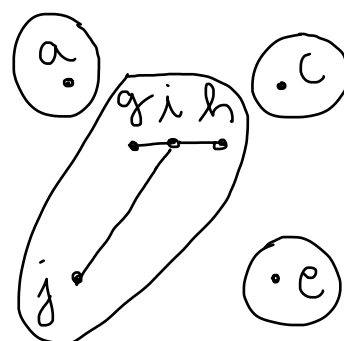
4. Podan je graf G .



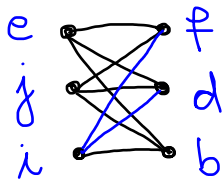
- (a) Izriši inducirani podgraf grafa G , ki je drevo na vsaj 5 vozliščih.
 (b) Ali je graf G Hamiltonov? Odgovor utemelji.
 (c) Ali je graf G ravninski? Odgovor utemelji.

(a) $a - f - g - i - h$

(b) NE; Odstranimo 3 vozlišča: b, f, d .
 Nastanejo 4 komponente.



(c) NE;



if : g
 id : h